RFR27 16-Channel Temperature Monitor New Member Project-10

Rutgers Formula Racing

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1 Revision Log

Editor	Date	Revision
Maaz Mansuri	Oct. 12 2026	Document Creation
Maaz Mansuri	Oct. 15 2026	Established outline of documentation and table of contents
Fayella Fenton	Oct. 15 2026	Inserted Source Refs

2 Purpose

Create at 16 channel temperature monitor and transmit data over I2C Bus

3 Parties Involved

Name	Team Role	Report Role
Maaz Mansuri	Software	Author
Fayella Fenton	Hardware	Author
Vaanika Singla	Electronics Lead	Reviewer
Jeevan Shah	Electronics Lead	Reviewer

4 Hardware Design & Signal Conditioning

5 Firmware and Software Design

6 Verification, Calibration, & Test Results

7 Bill of Materials

8 Sources

References

[] Understanding Our Components

[1] Robotics Back-End, *Arduino Uno - Pins Overview*, YouTube, 2021. Available: https://youtu.be/bniUECtJkeU

[2] Electronoobs, *Arduino Multiplexor - MUX INPUTS & OUTPUTS - ANALOG and PWM*, YouTube, 2021. Available: https://youtu.be/Dco6jo9xgAo

[3] Dejan, *How I2C Communication Works?*, HowToMechatronics. Available. https://howtomechatronics.com/tutorials/arduino/how-i2c-communication-works-and-how-to-use-it-with-arduino/

[4] CircuitBread, *Intro to Op-Amps (Operational Amplifiers) | Basic Circuits*, YouTube, 2022. Available: https://youtu.be/fmRHDqcodS4

[5] Temperature Sensor Datasheet. Available: https://enepaq.com/wp-content/uploads/2025/02/Sony-Murata-VTC5A-Li-ion-Battery-Module-With-Temperature-Sensor-Datasheet-.pdf

[] Circuit Basics

[6] RimstarOrg, *How to Read a Schematic*, YouTube, 2014. Available: https://youtu.be/_HZ-EQ8Hc8E

[7] EEVblog, *Playlist on DC Fundamentals*, YouTube, 2017. Available: https://youtu.be/_HZ-EQ8Hc8E

[] Arduino IDE